UNIT 2: SHAPES AND REGIONS

SESSION 3: SIZE FILTERING

SLO 1: APPLY SIZE-BASED FILTERING TECHNIQUES TO REMOVE NOISE AND IRRELEVANT OBJECTS FROM BINARY IMAGES.

Spot the Noise 🔍

In binary images, small isolated pixels or tiny components often represent noise, while larger components represent meaningful objects.

Observe the binary image below:

0 0 0 0 0 0

0 1 1 0 0 1

0 1 1 0 1 0

0 0 0 0 1 0

0 1 0 0 1 1

0 1 0 0 0 0

Task

1. **Circle or mark the small noisy components.**
2. **Identify the larger relevant objects.**

Count the total number of components before filtering.

Answer: • Answer: 3 components

- Apply Size Filtering (minimum size = 3 pixels)

Apply Size Filtering

Assume the minimum size threshold = 3 pixels.

Component	Number of Pixels	Keep / Remove
Object A	4	Keep
Object B	5	Keep
Object C	2	Remove
Object D	0	Remove

Questions

1. **Which components are considered irrelevant (noise)?**

Answer: Object C, because it contains only 2 pixels, which is below the minimum threshold of 3 pixels.

2. **Which components are relevant objects?**

Answer: Object A and Object B, because they contain 4 and 5 pixels respectively, both greater than or equal

3. **Total number of objects after filtering:**

Answer: 2 objects